

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Project Design Phase: Solution Architecture

Date:	October 3, 2026	Maximum Marks:	4 Marks
Team ID:	SWTID-2026-2585	Project Name:	Standard Laptop Automation

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

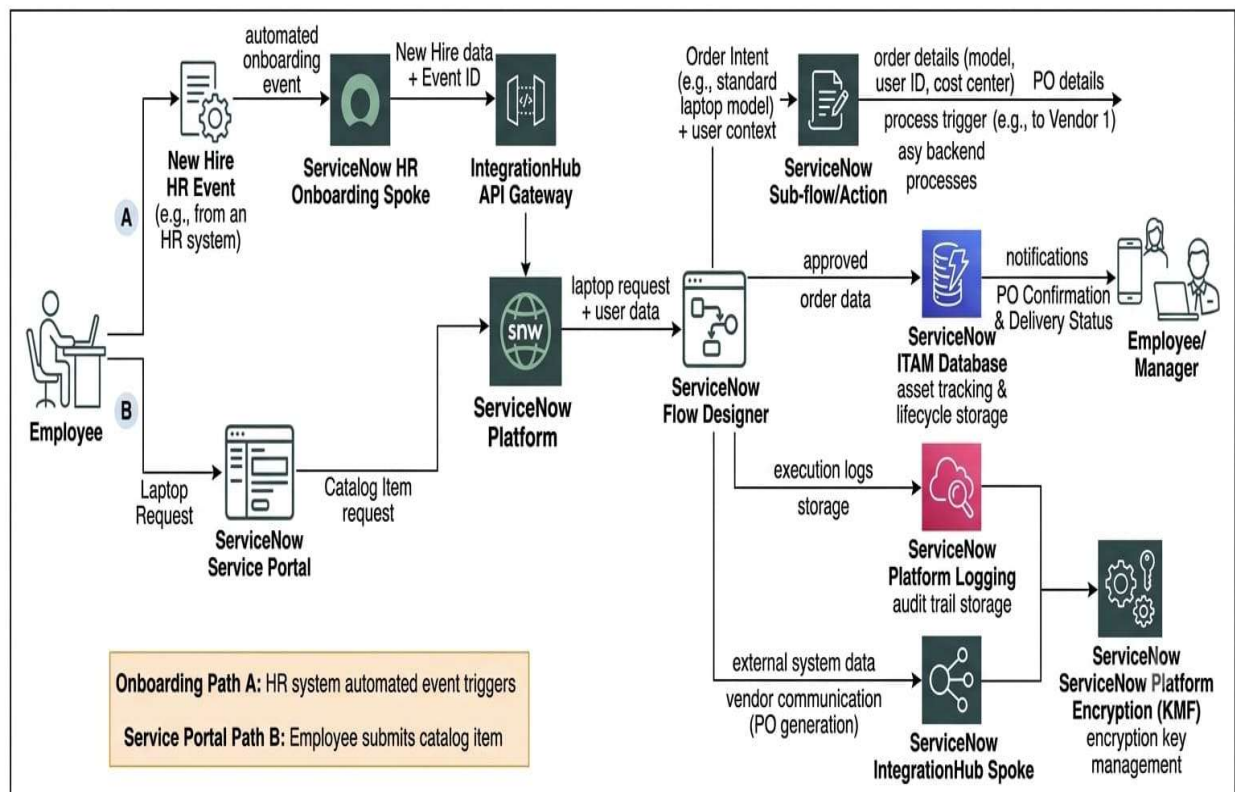


Figure 1: Architecture and data flow of the standard laptop automation sample application

Architecture components:

- The solution architecture for Streamlining IT Procurement using ServiceNow Flow Designer connects user touchpoints, enterprise automation layers, and backend asset management systems into a zero-touch workflow.
- Architecture Components & Flow
- User Interaction Layer (Frontend / Channels):
- ServiceNow Service Portal: Employees or managers select pre-approved laptop configurations from the Service Catalog.
- HR / Onboarding System Integration: Automated trigger from HR platforms (e.g., Workday) upon registering a new hire record.
- Collaboration Tools (MS Teams / Slack): Actionable notification cards pushed directly to managers for instant one-click approvals.
- Core Process & Automation Layer (ServiceNow Platform):
- Service Catalog & Request Management (sc_request / sc_req_item): Captures order parameters, user details, and delivery locations.
- ServiceNow Flow Designer (Workflow Engine): The central orchestration layer executing low-code logic, conditional branches, and parallel routing based on catalog inputs.
- Dynamic Approval Engine: Evaluates approval matrices (cost thresholds, department heads) and routes approval tasks automatically.
- Fulfillment & Integration Layer (Backend / Operations):
- IT Asset Management (ITAM) / Configuration Management Database (CMDB): Checks current stock levels and updates hardware inventory lifecycle statuses automatically.
- IntegrationHub (REST / SOAP Spoke): Triggers external REST API endpoints to vendor procurement portals (e.g., Dell, HP, Lenovo) for automated Purchase Order generation.
- Fulfillment Task Generation: Automatically assigns local IT warehouse tasks for physical staging, imaging, and courier dispatch if local stock is available.